Sunday
15 2 26

Assignment No 01

Q. 1 Print an n x n star square (n = 4)

→

```
class codex {
    Public static void main (String [] args) {
        for (int i = 1; i ≤ 4; i++) {
            for (int j = 1; j ≤ 4; j++) {
                System.out.print (" * ");
            }
            System.out.println ();
        }
    }
}
```

Dry Run →

① i = 1 , 1 ≤ 4 - true

↓

j = 1 , 1 ≤ 4 → sop("*") → j++ | *

j = 2 , 2 ≤ 4 → sop("*") → j++ | * *

j = 3 , 3 ≤ 4 → sop("*") → j++ | * * *

j = 4 , 4 ≤ 4 → sop("*") → j++ | * * * *

j = 5 , 5 ≤ 4 → false
└ sopln();

Inner for loop terminate
i Increamen होगा.

② $i = 2$, $2 \le 4$ – true

$j = 1$, $1 \le 4$
↳ sop("*")
j++

```
* * * *
*
```

$j = 2$, $2 \le 4$
↳ sop("*") → j++

```
* * * *
* *
```

$j = 3$, $3 \le 4$
↳ sop("*") → j++

```
* * * *
* * *
```

$j = 4$, $4 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
```

$j = 5$, $5 \le 5$ → false → sopln()

inner loop च्या बाहेर जाणार ; i increament होणार.

③ $i = 3$, $3 \le 4$ – true

$j = 1$, $1 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
*
```

$j = 2$, $2 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* *
```

$j = 3$, $3 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * *
```

$j = 4$, $4 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * * *
```

$j = 5$, $5 \le 4$ → false → sopln()

inner loop च्या बाहेर पडणार, i increment होणार.

④ $i = 4$, $4 \le 4 \rightarrow$ true

$j = 1$, $1 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * * *
*
```

$j = 2$, $2 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * * *
* *
```

$j = 3$, $3 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * * *
* * *
```

$j = 4$, $4 \le 4$
↳ sop("*") → j++

```
* * * *
* * * *
* * * *
* * * *
```

$j = 5$, $5 \le 4 \rightarrow$ false → sopln()

Inner loop च्या बाहेर पडणार आणि i increament होणार.

⑤ $i = 5$, $5 \le 4 \rightarrow$ false

outer loop मधुन बाहेर पडणार.

∴ o/p →

```
* * * *
* * * *
* * * *
* * * *
```

Q. 2 Print the following pattern

1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4

→

```
class codex {
    public static void main (string args[]) {
        for (int i = 1 ; i ≤ 4 ; i++) {
            for (int j = 1 ; j ≤ 4 ; j++) {
                System.out.print(i);
            }
            System.out.println();
        }
    }
}
```

Dry Run →

① i = 1 ; 1 ≤ 4 → true
↓

j = 1 , 1 ≤ 4 ↳ sop (i) → j++	1
j = 2 , 2 ≤ 4 ↳ sop (i) → j++	1 1
j = 3 , 3 ≤ 4 ↳ sop (i) → j++	1 1 1
j = 4 , 4 ≤ 4 ↳ sop (i) → j++	1 1 1 1

j = 5 , 5 ≤ 4 → False → sopln()

inner loop मधुन बाहेर पडणार आणि i increment होणार

② $i = 2$, $2 \le 4 \rightarrow$ true

↓

$j = 1$,	$1 \le 4$ ↳ sop(i) → j++	1 1 1 1 1
$j = 2$,	$2 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2
$j = 3$,	$3 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2
$j = 4$,	$4 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2 2

$j = 5$, $5 \le 4 \rightarrow$ false → sopln

Inner loop च्या बाहेर पडणार आणि i increment होणार.

③ $i = 3$, $3 \le 4 \rightarrow$ true

↓

$j = 1$,	$1 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2 2 3
$j = 2$,	$2 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2 2 3 3
$j = 3$,	$3 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2 2 3 3 3
$j = 4$,	$4 \le 4$ ↳ sop(i) → j++	1 1 1 1 2 2 2 2 3 3 3 3

$j = 5$, $5 \le 4 \rightarrow$ false → sopln()

inner loop च्या बाहेर पडणार आणि i increment होणार.

(4) $i = 4$, $4 \le 4 \rightarrow$ true

$j = 1$, $1 \le 4$
↳ sop(i) → j++

1	1	1	1
2	2	2	2
3	3	3	3
4			

$j = 2$, $2 \le 4$
↳ sop(i) → j++

1	1	1	1
2	2	2	2
3	3	3	3
4	4		

$j = 3$, $3 \le 4$
↳ sop(i) → j++

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	

$j = 4$, $4 \le 4$
↳ sop(i) → j++

1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4

$j = 5$, $5 \le 4 \rightarrow$ false
sopln()

Inner loop च्या बाहेर पडणार आणि i increment होणार

(5) $i = 5$, $5 \le 4$
↳ false

outer loop च्या पण बाहेर पडणार.

Q.3 Print the following pattern (n=4)

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

```
class codex {
    public static void main( String args [] ){
        for (int i = 1 ; i ≤ 4 ; i++) {
            for (int j = 1 ; j ≤ 4 ; j++) {
                System.out.print (j);
            }
            System.out.println ();
        }
    }
}
```

Dry Run →

① int i = 1 , 1 ≤ 4 → true
↓

j = 1 , 1 ≤ 4 ↳ sop (j) → j++	1
j = 2 , 2 ≤ 4 ↳ sop (j) → j++	1 2
j = 3 , 3 ≤ 4 ↳ sop (j) → j++	1 2 3
j = 4 , 4 ≤ 4 ↳ sop (j) → j++	1 2 3 4

j = 5 , 5 ≤ 4 → false → sopln()

inner loop च्या बाहेर पडणार आणि i increment होणार.

② $i = 2$; $2 \leq 4$ – true

$j = 1$, $1 \leq 4$
↳ sop (j) → j++

1	2	3	4
1			

$j = 2$, $2 \leq 4$
↳ sop(j) → j++

1	2	3	4
1	2		

$j = 3$, $3 \leq 4$
↳ sop (j) → j++

1	2	3	4
1	2	3	

$j = 4$, $4 \leq 4$
↳ sop (j) → j++

1	2	3	4
1	2	3	4

$j = 5$, $5 \leq 5$ → false → sopln

Inner loop च्या बाहेर पडणार व i increment होणार.

③ $i = 3$, $3 \leq 4$ → true

$j = 1$, $1 \leq 4$
↳ sop (j) → j++

1	2	3	4
1	2	3	4
1			

$j = 2$, $2 \leq 4$
↳ sop(j) → j++

1	2	3	4
1	2	3	4
1	2		

$j = 3$, $3 \leq 4$
↳ sop (j) → j++

1	2	3	4
1	2	3	4
1	2	3	

$j = 4$, $4 \leq 4$
↳ sop (j) → j++

1	2	3	4
1	2	3	4
1	2	3	4

$j = 5$, $5 \leq 4$ → false → sopln ()

Inner loop च्या बाहेर पडणार व i increment होणार

(4) $i = 4$, $4 \le 4$ — true

$j = 1$, $1 \le 4$
↳ sop (j) → j++

1 2 3 4
1 2 3 4
1 2 3 4
1

$j = 2$, $2 \le 4$
↳ sop (j) → j++

1 2 3 4
1 2 3 4
1 2 3 4
1 2

$j = 3$, $3 \le 4$
↳ sop(j) → j++

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3

$j = 4$, $4 \le 4$
↳ sop(j) → j++

1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4

$j = 5$, $5 \le 4$ → false → sopln()
Inner loop च्या बाहेर जाणार , i increment होणार

(5) $i = 5$, $5 \le 5$
↳ false

outer loop च्या बाहेर जाणार.